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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Euthanasia Agents: Mice were euthanized via anesthetic overdose using an intraperitoneal (IP) injection of avertin (2,2,2-tribromoethanol, Sigma-Aldrich), followed by cervical dislocation and decapitation for neurosphere assay and FACS experiments, or transcardial perfusion with 4 paraformaldehyde (PFA).
IACUC: All animal work was performed in accordance with institutional guidelines and approved by the University of Toronto Animal Care Committee.
Inclusion and Exclusion Criteria
Cells were first gated to exclude debris and doublets, followed by DAPI exclusion to gate for live cells.
Attrition
not detected.
Sex as a biological variable
Male and female mice were used for all experiments and were randomly allocated to experimental groups.
Subject Demographics
Age: Experimental model and subject details Animals Young adult (6-week-old) C57BL6 mice were purchased from Charles River (Montreal, Canada) for breeding.
Randomization
Male and female mice were used for all experiments and were randomly allocated to experimental groups.
Blinding
Quantification and statistical analysis Experimenters were blinded to groups.
Power Analysis
not detected.
Replication
not required.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies Rat monoclonal anti-CD133 (13A4), Biotinylated Thermo Fisher Cat 13-1331-82 RRID: AB_466591 Streptavidin PE-Cyanine7 Conjugate Thermo Fisher Cat 25-4317-82 RRID: AB_10116480 Mouse monoclonal anti-Doublecortin (E-6) Santa Cruz Biotechnology Cat sc-271390 RRID: AB_10610966 Rabbit polyclonal anti-NeuN Sigma-Aldrich Cat ABN78 RRID: AB_10807945 Rabbit polyclonal anti-Ki67 Abcam Cat ab15580 RRID: AB_443209 Goat polyclonal anti-AIF-1Iba1 Novus Biologicals Cat NB100-1028 RRID: AB_521594 Rabbit polyclonal anti-Iba1 Wako Chemicals Cat 019-19741 RRID: AB_839504 Rat monoclonal anti-mouse CD68 (FA-11) Bio-Rad Cat MCA1957 RRID: AB_322219 Donkey anti-mouse Alexa 568 Thermo Fisher Cat A10037 RRID: AB_2534013 Goat anti-rabbit Alexa 488 Thermo Fisher Cat A11034 RRID: AB_2576217 Donkey anti-rabbit Alexa 488 Thermo Fisher Cat A21206 RRID: AB_2535792 Donkey anti-goat Alexa 568 Thermo Fisher Cat A11057 RRID: AB_2534104 Donkey anti-rat Alexa 594 Thermo Fisher Cat A21209 RRID: AB_2535795 Chemicals, peptides, and recombinant proteins 2,2,2-tribromoethanol Sigma-Aldrich Cat T48402 Metformin (1,1-dimethylbiguanide hydrochloride) Sigma-Aldrich Cat D150959 DAPI Thermo Fisher Cat D1306 RRID: AB_2629482 EdU (5-ethynyl-2-deoxyuridine) Thermo Fisher Cat E10187 EGF-Alexa 647 Conjugate Thermo Fisher Cat E35351 Isoamyl acetate Sigma-Aldrich Cat W205508 Citralva (3,7-dimethyl-2,6-octadienenitrile) Sigma-Aldrich Cat W512508 Critical commercial assays Click-iT EdU Alexa 647	anti-CD133		
	Rat monoclonal anti-CD133	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# 13-1331-82, RRID:AB_466591)(link)
	Streptavidin	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# 25-4317-82, RRID:AB_10116480)(link)
	anti-Doublecortin	Santa Cruz Biotechnology	(Santa Cruz Biotechnology Cat# sc-271390, RRID:AB_10610966)(link)
	E-6		
	anti-NeuN		
	Rabbit polyclonal anti-NeuN	Millipore	(Millipore Cat# ABN78, RRID:AB_10807945)(link)
	anti-Ki67	Abcam	(Abcam Cat# ab15580, RRID:AB_443209)(link)
	anti-AIF-1Iba1		
	anti-AIF-1Iba1	Novus	(Novus Cat# NB 100-1028, RRID:AB_521594)(link)
	anti-Iba1		
	anti-Iba1	FUJIFILM Wako Pure Chemical Corporation	(FUJIFILM Wako Pure Chemical Corporation Cat# 019-19741, RRID:AB_839504)(link)
	anti-mouse CD68		
	anti-mouse	Bio-Rad	(Bio-Rad Cat# MCA1957, RRID:AB_322219)(link)
	anti-mouse		
	Donkey anti-mouse Alexa	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# A10037, RRID:AB_11180865)(link)
	anti-rabbit		
	Goat Alexa Cat	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# A-11034, RRID:AB_2576217)(link)
	Donkey Alexa	Molecular Probes	(Molecular Probes Cat# A-21206, RRID:AB_2535792)(link)
	anti-goat		
	Donkey Alexa Cat	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# A-11057, RRID:AB_2534104)(link)
	anti-rat		

Flow Cytometry Assay Kit Thermo Fisher Cat C10424 Experimental models: Organismsstrains C57BL6 Charles River Laboratories RRID: IMSR_CRL:27 FVBN-Tg(GFAPGFP)14MesJ The Jackson Laboratory RRID: IMSR_JAX:003257 Software and algorithms BD FACSDiva BD Biosciences https:www.bdbiosciences.comen-usinstrumentsresearch-instrumentsresearch-softwareflow-cytometry-acquisitionfacsdiva-software FlowJo 9.3 FlowJo LLC https:www.flowjo.com Zen Zeiss https:www.zeiss.commicroscopyintproductsmicroscope-softwarezen.html Fiji Schindelin et al. 81 https:imagej.netFiji Prism 6 GraphPad Software https:www.graphpad.comscientific-softwareprism Other 20mg sucrose pellets Bio-Serv Cat F05550 Resource availability Lead contact Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Cindi Morshead (cindi.morsheadutoronto.ca).	Donkey anti-rat Alexa Cat	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# A-21209, RRID:AB_2535795)(link)
	hydrochloride	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# D1306, RRID:AB_2629482)(link)
CD133 (prominin-1) was labeled using an anti-CD133-biotin antibody (Thermo Fisher, 13-1331-82) at a concentration of 1:300 in staining solution.	prominin-1		
	anti-CD133-biotin		
Neuroblasts were labeled using mouse anti-DCX (1:400 Santa Cruz Biotechnology, sc-271390), neurons were labeled using rabbit anti-NeuN (1:250 Sigma-Aldrich, ABN78), proliferating cells were labeled using rabbit anti-Ki67 (1:500 Abcam, ab15580), and microglia were labeled using goat anti-Iba1 (1:200 Novus Biologicals, NB100-1028) for Iba1+Ki67+ labeling (Figure 5E), or rabbit anti-Iba1 (1:500 Wako Chemicals, 019-19741) and rat anti-CD68 (1:500 Bio-Rad, MCA1957) for Iba1+CD68+ labeling (Figure 5F) .	anti-Ki67		
The next day, slides were incubated in secondary antibodies (1:750) and 1:10,000 DAPI (Thermo Fisher, D1306) for 90 minutes at room temperature prior to applying	S302380-2		

mounting medium (Dako, S302380-2) and coverslips.			
Experimental Models: Organisms/Strains			
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies Rat monoclonal anti-CD133 (13A4), Biotinylated Thermo Fisher Cat 13-1331-82 RRID: AB_466591 Streptavidin PE-Cyanine7 Conjugate Thermo Fisher Cat 25-4317-82 RRID: AB_10116480 Mouse monoclonal anti-Doublecortin (E-6) Santa Cruz Biotechnology Cat sc-271390 RRID: AB_10610966 Rabbit polyclonal anti-NeuN Sigma-Aldrich Cat ABN78 RRID: AB_10807945 Rabbit polyclonal anti-Ki67 Abcam Cat ab15580 RRID: AB_443209 Goat polyclonal anti-AIF-1Iba1 Novus Biologicals Cat NB100-1028 RRID: AB_521594 Rabbit polyclonal anti-Iba1 Wako Chemicals Cat 019-19741 RRID: AB_839504 Rat monoclonal anti-mouse CD68 (FA-11) Bio-Rad Cat MCA1957 RRID: AB_322219 Donkey anti-mouse Alexa 568 Thermo Fisher Cat A10037 RRID: AB_2534013 Goat anti-rabbit Alexa 488 Thermo Fisher Cat A11034 RRID: AB_2576217 Donkey anti-rabbit Alexa 488 Thermo Fisher Cat A21206 RRID: AB_2535792 Donkey anti-goat Alexa 568 Thermo Fisher Cat A11057 RRID: AB_2534104 Donkey anti-rat Alexa 594 Thermo Fisher Cat A21209 RRID: AB_2535795 Chemicals, peptides, and recombinant proteins 2,2,2-tribromoethanol Sigma-Aldrich Cat T48402 Metformin (1,1-dimethylbiguanide hydrochloride) Sigma-Aldrich Cat D150959 DAPI Thermo Fisher Cat D1306 RRID: AB_2629482 EdU (5-ethynyl-2-deoxyuridine) Thermo Fisher Cat E10187 EGF-Alexa 647 Conjugate Thermo Fisher Cat E35351 Isoamyl acetate Sigma-Aldrich Cat W205508 Citralva (3,7-dimethyl-2,6-octadienenitrile) Sigma-Aldrich Cat W512508 Critical commercial assays Click-iT EdU Alexa 647 Flow Cytometry Assay Kit Thermo Fisher Cat C10424 Experimental	C57BL6		RRID:IMSR_CRL:027(link)
	FVBN-Tg(GFAPGFP)14Mes J The		RRID:IMSR_JAX:003257(link)

models: Organismsstrains C57BL6 Charles River Laboratories RRID: IMSR_CRL:27 FVBN-Tg(GFAPGFP)14MesJ The Jackson Laboratory RRID: IMSR_JAX:003257 Software and algorithms BD FACSDiva BD Biosciences https:www.bdbiosciences.comen-usinstrumentsresearch-instrumentsresearch-softwareflow-cytometry-acquisitionfacsdiva-software FlowJo 9.3 FlowJo LLC https:www.flowjo.com Zen Zeiss https:www.zeiss.commicroscopyintproductsmicroscope-softwarezen.html Fiji Schindelin et al. 81 https:imagej.netFiji Prism 6 GraphPad Software https:www.graphpad.comscientific-softwareprism Other 20mg sucrose pellets Bio-Serv Cat F05550 Resource availability Lead contact Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Cindi Morshead (cindi.morsheadutoronto.ca).			
Experimental model and subject details Animals Young adult (6-week-old) C57BL6 mice were purchased from Charles River (Montreal, Canada) for breeding.	C57BL6		Suggestion: RRID:IMSR_JAX:000664(link)
Fluorescence-activated cell sorting (FACS) GFAP::GFP mice were sacrificed on P19 and the SEZ tissue from 5 mice per treatment group was pooled in order to obtain a sufficient number of SEZ cells per FACS run as previously described.	GFAP::GFP		
Software and Algorithms			
STARMethods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies Rat monoclonal anti-CD133 (13A4), Biotinylated Thermo Fisher Cat 13-1331-82 RRID: AB_466591 Streptavidin PE-Cyanine7 Conjugate Thermo Fisher Cat 25-4317-82 RRID: AB_10116480 Mouse monoclonal anti-Doublecortin (E-6) Santa Cruz Biotechnology Cat sc-271390 RRID: AB_10610966 Rabbit polyclonal anti-NeuN Sigma-Aldrich Cat ABN78 RRID: AB_10807945 Rabbit polyclonal anti-Ki67 Abcam Cat ab15580 RRID:	BD FACSDiva BD Biosciences		
	Prism		Suggestion: Meta-Prism (RRID:SCR_021836)(link)
	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

AB_443209 Goat polyclonal anti-AIF-1Iba1 Novus Biologicals Cat NB100-1028 RRID: AB_521594 Rabbit polyclonal anti-Iba1 Wako Chemicals Cat 019-19741 RRID: AB_839504 Rat monoclonal anti-mouse CD68 (FA-11) Bio-Rad Cat MCA1957 RRID: AB_322219 Donkey anti-mouse Alexa 568 Thermo Fisher Cat A10037 RRID: AB_2534013 Goat anti-rabbit Alexa 488 Thermo Fisher Cat A11034 RRID: AB_2576217 Donkey anti-rabbit Alexa 488 Thermo Fisher Cat A21206 RRID: AB_2535792 Donkey anti-goat Alexa 568 Thermo Fisher Cat A11057 RRID: AB_2534104 Donkey anti-rat Alexa 594 Thermo Fisher Cat A21209 RRID: AB_2535795 Chemicals, peptides, and recombinant proteins 2,2,2-tribromoethanol Sigma-Aldrich Cat T48402 Metformin (1,1-dimethylbiguanide hydrochloride) Sigma-Aldrich Cat D150959 DAPI Thermo Fisher Cat D1306 RRID: AB_2629482 EdU (5-ethynyl-2-deoxyuridine) Thermo Fisher Cat E10187 EGF-Alexa 647 Conjugate Thermo Fisher Cat E35351 Isoamyl acetate Sigma-Aldrich Cat W205508 Citralva (3,7-dimethyl-2,6-octadienenitrile) Sigma-Aldrich Cat W512508 Critical commercial assays Click-iT EdU Alexa 647 Flow Cytometry Assay Kit Thermo Fisher Cat C10424 Experimental models: Organismsstrains C57BL6 Charles River Laboratories RRID: IMSR_CRL:27 FVBN-Tg(GFAPGFP)14MesJ The Jackson Laboratory RRID: IMSR_JAX:003257 Software and algorithms BD FACSDiva BD Biosciences https:www.bdbiosciences.comen-usinstrumentsresearch-instrumentsresearch-softwareflow-cytometry-acquisitionfacsdiva-software FlowJo 9.3 FlowJo LLC https:www.flowjo.com Zen Zeiss https:www.zeiss.commicroscopyintproductsmicroscope-softwarezen.html Fiji Schindelin et al. 81 https:imagej.netFiji Prism 6 GraphPad Software https:www.graphpad.comscientific-softwareprism Other 20mg sucrose				

pellets Bio-Serv Cat F05550 Resource availability Lead contact Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Cindi Morshead (cindi.morsheadutoronto.ca).			
The supernatant was replaced with DMEMF12 (DMEM from GIBCO, 12100046 F12 from GIBCO, 21700075), which was supplemented with ovomucoid solution (7mgmL Worthington, LS003086) and DNase (15mgmL Worthington, LK003172).	GIBCO		
BD FACSDiva software (BD Biosciences, California, USA) was used for equipment setup and data acquisition, while FlowJo 9.3 (FlowJo LLC, Oregon, USA) was used for data analysis.	BD FACSDiva		
	FlowJo		Suggestion: FlowJo, RRID:SCR_008520(link)
Images of the SEZ (Figure 3B) were tiled upon acquisition using Zen software with 10 overlap between tiles in order to capture the entire lateral ventricle.	Zen		
All cells were manually counted using the Fiji Cell Counter plugin.	Fiji		
All raw data was recorded using Microsoft Excel (Microsoft, Washington, USA).	Microsoft Excel		Suggestion: Microsoft Excel, RRID:SCR_016137(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Column analyses between 2 groups were performed using unpaired t tests, or multiple unpaired t tests when comparing 2 groups over multiple time points.	t tests
Column analyses between 3 groups were performed using one-way ANOVA with Tukeys test for multiple comparisons.	ANOVA
	Tukeys test

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

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